

1. What is this presentation trying to explain?

Your instructor wants you to understand **why LLMs (Large Language Models) exist, why they were built this way, what problems they solve, and what new problems they create.**

This is not a full NLP course — it's the story of how LLMs evolved.

2. What is a “word”? Why is it complicated?

We humans think “words” are simple:

“The cat sat on the mat” — 6 words.

But computers can't rely on that because:

a) Spaces don't define words in every language

- English: uses spaces
- Chinese: *no spaces* between words
- German: combines many words into one giant word
 - e.g., **Donaudampfschiffahrtsgesellschaftskapitän** (one word!)

So the idea of a “word” is not universal.

b) Orthographic vs grammatical words

Slide example:

I'm

- Orthographically: one word
- Grammatically: “I am” → two words

So “word” is not a stable, consistent unit.

3. The “Vocabulary Problem”

A language’s vocabulary is:

- Open-ended
- Always growing
- New slang, new names, new technical terms, new brands

A computer cannot store *every* possible word.

Even if you store 1 million words, the next day someone invents a new one.

So...

LLMs DO NOT USE WORDS

They use **subwords**.

4. Subwords and BPE (Byte Pair Encoding)

Subwords = small pieces of words.

Examples:

- “playing” → play + ing

- “unhappiness” → un + happi + ness
- “Donaudampfschiffahrt...” → broken into small parts

Benefits:

- Handles **unknown words**
- Breaks long words into learnable pieces
- Works for all languages

This solves the vocabulary problem.

5. Tokens

A token = smallest unit the model uses.

It might be:

- a whole word
- a subword
- a single character
- even punctuation

Different models tokenize differently.

Why tokens matter:

- Models don't read text; they read **numbers**
 - Tokens → converted into **embeddings** → then processed
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6. Embeddings (the meaning of words as numbers)

Every token is turned into a **vector**: a list of numbers in high-dimensional space.

Important idea:

Tokens with similar meaning end up close to each other in vector space.

Example:

- “king” and “queen” are close
- “walk” and “run” are close
- “banana” is far from “democracy”

Embeddings allow math to capture meaning.

How is this learned?

Through training — not manually.

7. Why RNNs failed

Before Transformers, models used RNNs (Recurrent Neural Networks).

They had two major problems:

a) Vanishing gradient problem

When sequences are long, information from earlier words “fades away.”

The model forgets context.

b) Slow

RNNs process tokens **one by one** (sequentially).

You cannot parallelize.

This makes training extremely slow.

8. Transformers — the major breakthrough

Transformers solved **both** problems.

They have *no recurrence*.

Instead, they introduced the idea of **attention**.

Attention = the model deciding “which words matter” at each step.

Transformers process all tokens **in parallel**, which is thousands of times faster.

Architecture:

- **Encoder**: reads/understands
- **Decoder**: generates text
- **Encoder–Decoder**: does sequence-to-sequence tasks (translation, summarization)

9. Positional Encoding

Since Transformers don't read tokens in order,
they need a way to know **the position** of each word.

Positional encoding = give each token a number representing its order.

This allows:

- “dog bites man”
- “man bites dog”

...to be understood differently.

10. Attention (Q, K, V)

This is the heart of Transformers.

Every token in a sentence gets three vectors:

- **Q (query)**: what am I looking for?
- **K (key)**: how relevant am I to others?
- **V (value)**: what information do I carry?

The model computes:

$$\text{Attention score} = Q \cdot K$$

Then applies softmax $\rightarrow$ weights $\rightarrow$ weighted sum of V.

This is how the model decides:

- Which tokens matter
 - How much each token influences the output
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11. Self-Attention

Self-attention lets each token look at *every other token* in the sentence.

Example:

“The dog that chased the cat was hungry.”

The model can relate:

- “dog” $\leftrightarrow$ “was hungry”
- “that” $\leftrightarrow$ “chased”
- “chased” $\leftrightarrow$ “cat”

This gives global context.

12. Multi-Head Attention

Self-attention is done **multiple times in parallel**.

Each “head” focuses on different patterns:

- Head 1: grammar relationships
- Head 2: semantic meaning
- Head 3: positional relations
- Head 4: verb-object relations
- etc.

Combining multiple heads gives richer understanding.

13. Types of Transformer Models

Encoder-only (BERT)

- Understand text
- Classification, NER, embeddings
- Not used for text generation

Decoder-only (GPT)

- Generates text
- Autoregressive

- Good for writing, chatting, continuing text

Encoder–Decoder (T5, BART)

- Translation
 - Summarization
 - Q/A
 - Any “input → output” task
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14. What is a Language Model?

A language model predicts:

$$p(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3 \dots \mathbf{x}_n)$$

The probability of a sequence of tokens.

It generates one token at a time:

$$\text{next_token} = f(\text{previous_tokens})$$

This is called **autoregressive** generation.

15. Self-supervised learning

The magic of LLMs:

- They don’t need human-labeled data.
- They learn by predicting missing tokens in text.

Example:

“I ____ to the store.”

Model learns to predict “went.”

Every sentence on the internet becomes training data automatically.

16. Do LLMs “understand”?

Your teacher gives three viewpoints:

YES → Emergent Abilities

When models get big enough, they show unexpected skills:

- reasoning
- analogies
- coding
- creativity

These abilities **emerge** without being explicitly trained.

NO → Stochastic Parrots

Another view says:

- LLMs don’t think
- They just mimic statistical patterns
- They only repeat what sounds right
- They don’t have real understanding

HALLUCINATIONS

Models sometimes:

- Make up facts

- Produce incorrect information
Why?
Because they are probabilistic text generators.

IDK → Tesler's Law

“AI is whatever hasn't been done yet.”

Meaning:

- When AI achieves something, people no longer think it's “intelligent.”
 - Understanding keeps evolving.
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17. Core Conclusion

LLMs:

- cannot rely on words
- use subwords (tokens)
- store meaning in embeddings
- use attention to relate words
- use multi-head attention for rich understanding
- use positional encoding for order
- are built from encoder/decoder blocks
- are trained using self-supervision
- can show emergent intelligence
- can also hallucinate

- raise philosophical questions about “understanding”

This is the entire story.

1. What is the core reason NLP systems avoid using “words” as the basic unit of tokenization?

- A. Words are too meaning-rich
- B. Words vary too much across speakers
- C. The vocabulary of any language is unbounded
- D. Tokens must always be characters

Answer: C

2. Why can't we just store a larger and larger vocabulary to solve the unknown-word problem?

- A. Models cannot store more than 50k words
- B. More data always reduces performance
- C. Vocabulary of natural language is open-ended
- D. Subword models require fixed vocabulary

Answer: C

3. Subword tokenization (like BPE) primarily helps with:

- A. Improving grammar of generated text
- B. Eliminating unknown tokens
- C. Ensuring syntactic correctness
- D. Reducing embedding dimension

Answer: B

4. In Transformer models, positional information is handled by:

- A. Recurrence
- B. Convolution
- C. Learned or sinusoidal positional encodings
- D. Token-level RNN gates

Answer: C

5. Why did RNNs fail to scale for large-context language modeling?

- A. They weren't differentiable
- B. They had vanishing gradients and sequential processing
- C. They couldn't perform classification
- D. They used too few parameters

Answer: B

6. Which of the following is NOT a component of attention computation?

- A. Query
- B. Key
- C. Value
- D. Gate

Answer: D

7. In self-attention, tokens attend to:

- A. Only the next token
- B. All tokens in the sequence

- C. Only previous tokens
- D. Only syntactically similar tokens

Answer: B

8. Masked attention is used in:

- A. Encoder
- B. Decoder
- C. Embedding layer
- D. Only during evaluation

Answer: B

9. The output of attention for a query is:

- A. A weighted sum of key vectors
- B. A weighted sum of value vectors
- C. A concatenation of keys and queries
- D. A dot-product of queries and values

Answer: B

10. Why does multi-head attention work better than single-head attention?

- A. It reduces parameter count
- B. It allows parallel positional encodings
- C. Each head learns different relational patterns
- D. It speeds up training by removing softmax

Answer: C

11. Embeddings work because:

- A. They compress tokens into one-hot vectors
- B. Words with similar meaning form similar vectors
- C. They remove semantic information intentionally
- D. They can only represent characters, not subwords

Answer: B

12. Transformer encoders are best suited for:

- A. Text generation
- B. Named entity recognition
- C. Autoregressive token prediction
- D. Sequence prediction without input

Answer: B

13. Decoder-only models like GPT are optimized for:

- A. Reasoning classification tasks
- B. Denoising corrupted sequences
- C. Autoregressive generation
- D. Clustering embeddings

Answer: C

14. Sequence-to-sequence transformer models (T5/BART) are ideal for:

- A. Auto-complete tasks
- B. Summarization and translation
- C. Sentence classification
- D. Static embedding creation

Answer: B

15. A language model defines:

- A. A rule-based grammar
- B. A probability distribution over token sequences
- C. A set of hard-coded templates
- D. A structured syntax parser

Answer: B

16. Language models are trained using:

- A. Manual annotation
- B. Supervised labels
- C. Self-supervision
- D. Reinforcement

Answer: C

17. The probability calculation in autoregressive LMs relies on:

- A. Joint probability of all future tokens
- B. Conditional probability of next token given previous ones
- C. Independent random sampling
- D. Hidden lexicon lookups

Answer: B

18. The main reason attention improves over RNNs is:

- A. It increases the vocabulary size
- B. It allows direct access to every token in the sequence
- C. It removes training data requirements
- D. It replaces embeddings

Answer: B

19. Why is the transformer architecture massively parallelizable?

- A. It removes neural networks entirely
- B. All tokens pass through layers independently
- C. Keys and values are precomputed with RNNs
- D. Positional encoding eliminates matrix multiplications

Answer: B

20. In attention, softmax is used to:

- A. Normalize similarities into probability weights
- B. Compute positional encoding
- C. Reduce dimensionality
- D. Select important tokens deterministically

Answer: A

21. "A word is known by the company it keeps" in embeddings refers to:

- A. Syntax-first modeling
- B. Co-occurrence based semantic similarity
- C. Attention masking
- D. BPE merge rules

Answer: B

22. Why can Chinese not rely on whitespace tokenization?

- A. The language has no punctuation
- B. Each character already represents a morpheme
- C. Characters are too long
- D. Chinese uses Latin scripts

Answer: B

23. The German compound word example in the slides demonstrates:

- A. Why morphemes must be ignored
- B. Why whitespace tokenization fails
- C. Why characters are always optimal tokens
- D. That German has fixed vocabulary sizes

Answer: B

24. Emergent abilities in LLMs refer to:

- A. Capabilities explicitly programmed
- B. Abilities that arise only at large scale
- C. Model hallucinations
- D. Token-level memorization

Answer: B

25. The term "stochastic parrot" criticizes LLMs because they:

- A. Generate only random outputs
- B. Lack true understanding and repeat patterns
- C. Use too many parameters
- D. Cannot generate long sequences

Answer: B

26. Hallucinations in LLMs occur because:

- A. Softmax always chooses the correct token
- B. The model fills uncertain gaps with plausible tokens
- C. Transformer attention fails
- D. Training data contains only fiction

Answer: B

27. Positional encodings help the Transformer with:

- A. Memory storage
- B. Sequential order awareness
- C. Reducing embeddings
- D. Creating attention masks

Answer: B

28. Which model type cannot generate text unless given an input sequence?

- A. Encoder-only
- B. Decoder-only
- C. Encoder-decoder
- D. All transformers can generate text

Answer: A

29. The $Q \cdot K^T$ similarity score measures:

- A. How important a value vector is
- B. How relevant one token is to another
- C. The position of tokens
- D. The softmax output directly

Answer: B

30. The reason self-supervision works well for LLMs is:

- A. Tokens naturally contain their own labels
- B. Labels must be manually generated
- C. It reduces context windows
- D. It requires no embeddings

Answer: A